

Kert Laansalu

19 Silverton Road, London, W6 9NY
+372 5166291, kertlaansalu@gmail.com, <https://kert114.github.io/>

EDUCATION

Imperial College London

London, United Kingdom

Integrated Master's of Electronic and Computer Engineering, MEng

2020 – 2024

- Graduated with 1st class honours
- Thesis: Representation Learning in Decentralised Multi-Agent Systems
- Modules: Deep Learning, Machine Learning, Robotics, Compilers, Advanced Programming

WORK & LEADERSHIP EXPERIENCE

Estonian Cyberforce - (*Rust, DustDDS, C++, Ansible, Networking*)

Tallinn, Estonia

Integrations Software Engineer

June 2025 – Present

- Integrated a laser range finder targeting system into Estonia's command-and-control infrastructure
- Developed distributed communication services in Rust, using DDS to support real-time transmission of targeting and sensor data. Built reusable integration components that improved platform scalability

Valent Projects - (*React, Python, GraphQL, C++*)

London, UK

Software Engineer

September 2024 – June 2025

- Led development of an end-to-end intelligence reporting platform that analysed millions of media articles and generated automated client-specific reports on sentiment, narrative evolution, and reputational risk
- Built ML-powered analysis workflows to evaluate sentiment, factual consistency, narrative trends, and reputational outlook, transforming unstructured media content into actionable business insights

SEB Bank - (*Java, Spring Boot, Angular*)

Tallinn, Estonia

Software Engineering Placement

April – September 2023

- Designed, developed, and deployed an end-to-end incentives automation system that triggered personalised customer email campaigns to drive Apple Pay onboarding and completion rates
- Built backend services using Java and Spring Boot to orchestrate incentive delivery logic, event handling, and campaign execution across banking systems

PROJECTS & PROFESSIONAL EXPERIENCE

Representation Learning for Collective Intelligence

London, UK

Master's thesis in collaboration with Imperial College London

2023 – 2024

- Developed a decentralised multi-agent SimCLR-based representation learning framework to study collaborative learning under non-IID data distributions, improving robustness and generalisation over large single-agent models
- Performed comparative analysis of Exact Diffusion and classical diffusion methods, evaluating convergence behaviour and stability in distributed learning settings

Avatour – (*C++, ROS, Verilog, HTML, JavaScript, SQL, PHP, AWS*)

- Led development of a telepresence robotics platform enabling remote operation of Boston Dynamics Spot via 360° video streaming and VR-based control interfaces, supporting immersive situational awareness for remote users
- Designed and implemented an integration layer connecting real-time video streaming pipelines with robot control systems, enabling low-latency, stable remote operation and synchronized perception-action feedback

Autonomous Mars Rover Project – (*C++, Verilog, HTML, JavaScript, SQL, PHP, AWS*)

- Developed an autonomous Mars rover system using ESP32 and FPGA (DE10-Lite) capable of environmental exploration, obstacle detection and avoidance, data logging, and remote communication
- Integrated computer vision and tracking modules with a multi-mode control architecture supporting remote control, autonomous navigation, and shortest-path routing to target coordinates

Augmented Reality IoT project – (*Python, Verilog, React Native, NoSQL, AWS*)

- Developed an augmented reality IoT application that overlays real-time navigation instructions, weather data, and nearby user locations within an interactive AR interface
- Designed and implemented bidirectional communication between a handheld FPGA controller and the mobile application over TCP, enabling real-time command execution and cloud-synced data retrieval

C90 to MIPS compiler – (*C++, Flex, Bison*)

- Developed a C90-to-MIPS compiler using C++, Flex, and Bison that translates pre-processed C90 into MIPS assembly, implementing full support for core language constructs like control flow, functions, and variable handling

SKILLS & ACHIEVEMENTS

Technical Skills: Java, Python, C++, Rust, React Native, Node.js, Spring Boot, SQL, AWS, Git, Verilog, Flex

Languages: English (native), Estonian (native), French (proficient), Russian (moderate)

Interests: Robotics, Choir, Debating, Folk dancing, Skiing, Football